AMITY UNIVERSITY KOLKATA

Weekly Progress Report

For the Week Commencing 12 October 2026

Enrollment No. A914138123021

NAME: UJAN SEN

Program: B.Sc Data Science Honors/Research

Company Name :
(Please mention if Applicable)

Faculty Guide's Name : DR. ANWESHA LAW

Industry Guide's Name : Not Applicable
(Please mention if Applicable)

Internship Mode : Offline

Project Title:
Route-Aware Machine Learning Framework for GPS-Based ETA Prediction

Targets for the week:
Complete work for: Route snapping and reference polylines. Implemented snapping of recorded points to representative morning and evening route polylines.

Achievements for the week

Calculated nearest-route positions and off-route distances so noisy coordinates can be compared on a common route.

Future work plans

Continue the next pipeline stage, validate the outputs, and document decisions for the project report.

Week's Summary

Days / Time	
Monday	12 Oct: Selected reference sessions for both directions. The reference route provides a common geometry for both commute directions.
Tuesday	13 Oct: Projected GPS points onto the reference polylines. Each point receives a nearest position on that geometry.
Wednesday	14 Oct: Calculated along-route distance. The distance measure identifies observations that drift away from the route.
Thursday	15 Oct: Measured off-route error. The projected progress can be compared across different sessions.
Friday	16 Oct: Reviewed the snapped CSV outputs. The generated snapped files preserve the original session identity.